PROJECT REPORT

Machine Health Monitoring and Anomaly Detection using Power BI

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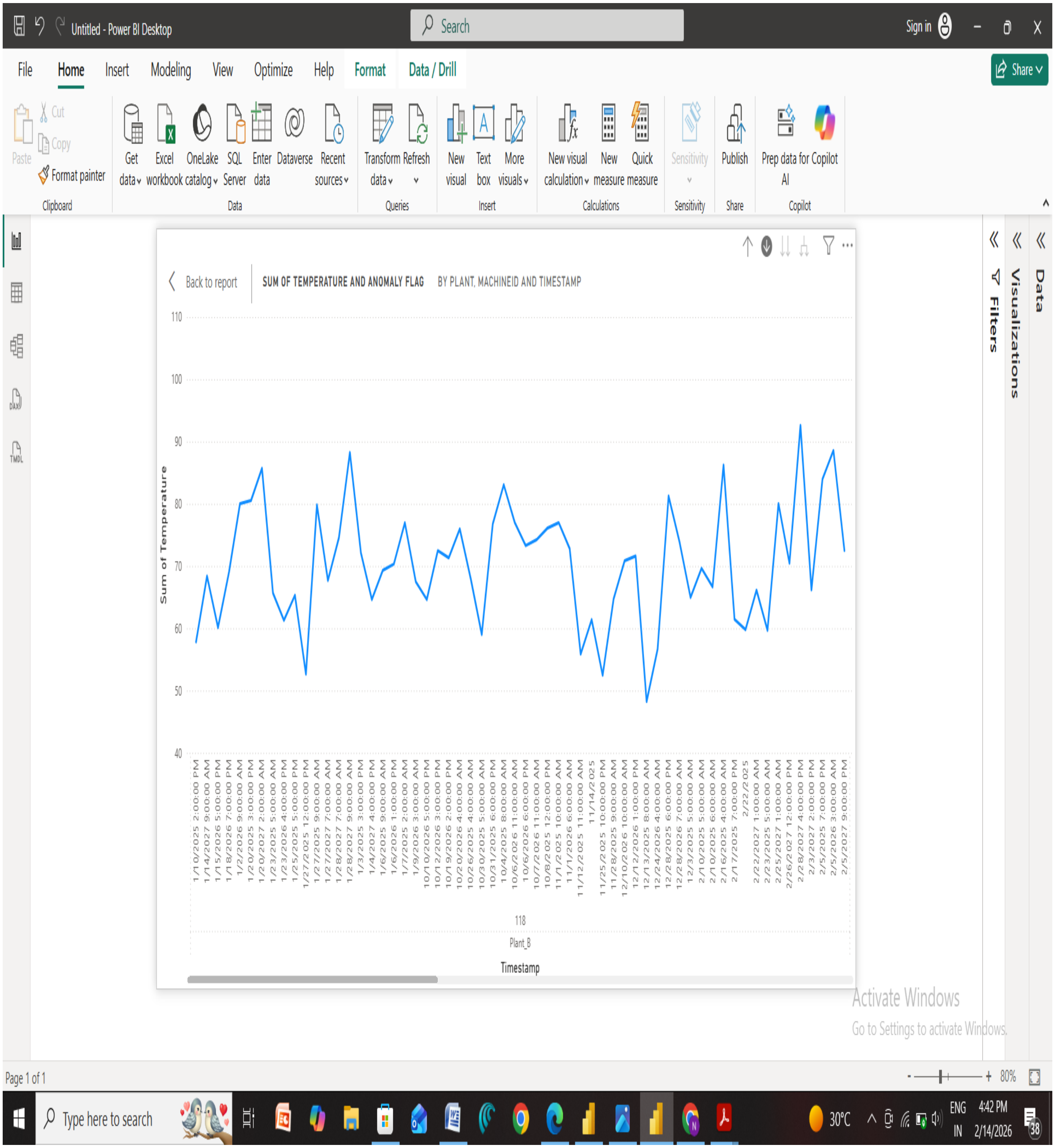

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> Machine Health Monitoring — Project Explanation

1.Introduction

This project focuses on monitoring machine health using operational data such as temperature, vibration, pressure, and energy consumption. The goal is to detect abnormal behavior (anomalies) that may indicate potential machine faults or performance issues. Power BI is used as the analytics and visualization tool to analyze trends and highlight unusual patterns.

Predictive maintenance is important in industries because early detection of faults helps reduce downtime, maintenance cost, and production loss. By analyzing temperature variations over time, this project demonstrates how anomalies can be identified using data analytics techniques.

2.Objective

The main objectives of the project are:

- To analyze machine operational data across different plants
- To visualize temperature trends using dashboards
- To detect abnormal temperature spikes using DAX calculations
- To enable drill-down analysis for deeper investigation
- To support decision-making for maintenance planning

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3. Dataset Description

The dataset contains machine monitoring information with the following fields:

- Plant — Location of machine
- MachineID — Unique machine identifier
- Timestamp — Date and time of reading

- Temperature — Machine temperature
- Pressure — Operating pressure
- Vibration — Machine vibration level
- Energy Consumption — Power usage
- Defect Count — Number of defects observed
- Maintenance Flag — Indicates maintenance activity

These variables help understand machine behavior over time.

4.Methodology

Step 1 — Load Data

The dataset is imported into Power BI using the "Get Data" option. Data types are verified to ensure timestamps and numeric values are correct.

Step 2 — Create Hierarchy

A hierarchy is created:

▯ Plant → MachineID → Timestamp

This allows drill-down analysis from overall plant level to individual machine readings.

Step 3 — Build Visualizations

Line charts are created to show temperature trends over time. Filters are applied to analyze specific machines or plants.

Step 4 — Create Anomaly Detection Measure

A DAX formula compares current average temperature with overall average temperature. If temperature exceeds a threshold (e.g., 20% higher), it is flagged as an anomaly.

Example logic:

- If temperature > normal threshold → Anomaly = 1
- Else → Anomaly = 0

Step 5 — Add Tooltips and Filters

The anomaly flag is added to tooltips so users can easily see abnormal points while hovering over charts.

5.Visualization and Analysis

The dashboard shows how temperature changes across time and machines. Users can drill down to investigate specific readings. Sudden spikes or drops indicate abnormal behavior.

Interactive filters allow selecting a plant or machine to study performance patterns. This helps identify which machine needs attention.

6.Results

The project successfully demonstrates:

- Detection of abnormal temperature readings
- Clear visualization of machine trends
- Easy navigation using drill-down
- Data-driven insights for maintenance

Anomaly flags highlight potential issues before failures occur.

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7. Advantages

- Improves predictive maintenance

- Reduces unexpected downtime
- Easy to interpret dashboards
- Scalable for large datasets
- Supports real-time monitoring

8.Limitations

- Uses simple threshold logic (not advanced ML)
- Depends on data quality
- Does not include real sensor streaming
- May require tuning thresholds

9.Future Enhancements

- Integrate machine learning for smarter anomaly detection
- Connect to real-time IoT sensors
- Add alerts for critical conditions
- Include predictive failure modeling
- Build automated reporting

>Conclusion

This project demonstrates how Power BI can be used to monitor machine health and detect anomalies using operational data. By visualizing temperature trends and applying DAX calculations, abnormal behavior can be identified early. Such analytics solutions improve maintenance efficiency and support proactive decision-making in industrial environments.